



Faculty of Science –Menofia University
Mathematics and Computer Science



Computer Graphics

M 4310

Chapter 2

2D Transformation

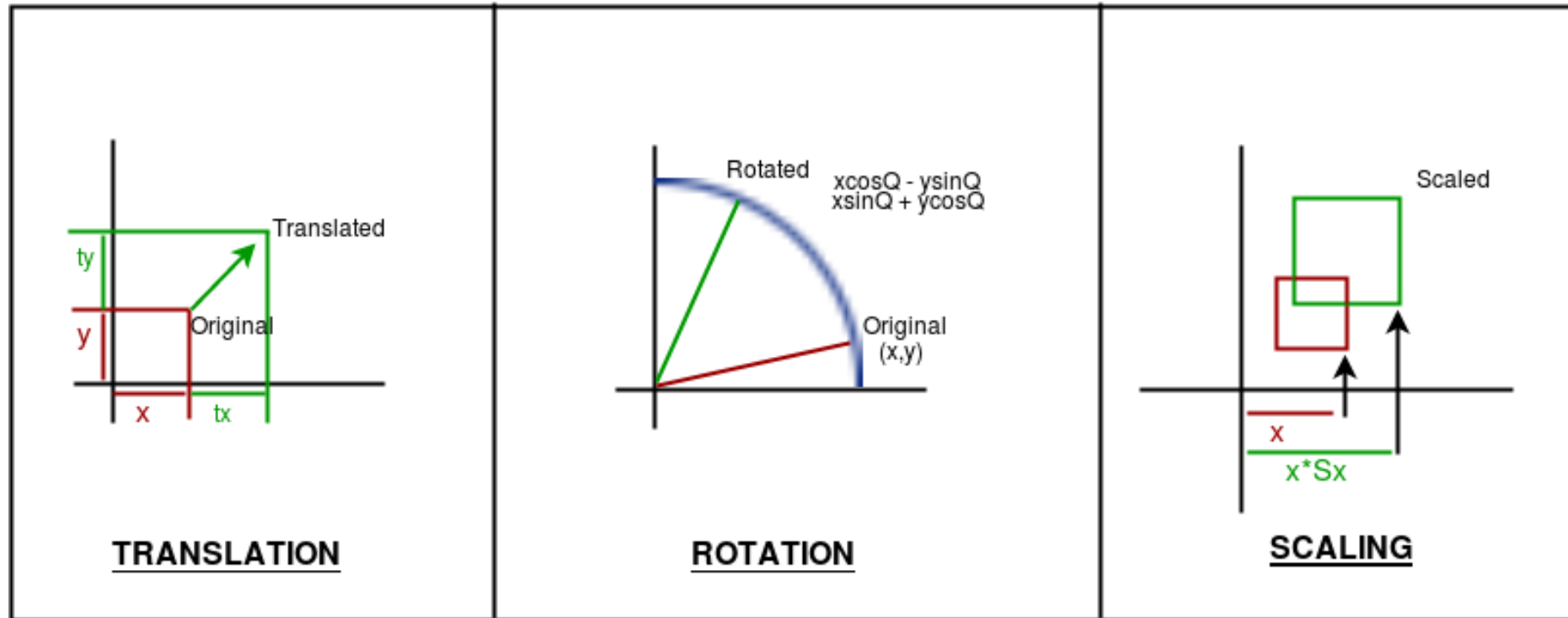
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Agenda

- Introduction to 2d Transformation
- Basic 2D transformation
- Matrix Representation
- Windows to viewport transformation

Transformation

1. A **transformation** is a mapping from one coordinate space to another that changes the geometric properties of graphical objects while preserving their basic structure.
2. A **transformation** in computer graphics is a mathematical operation that changes the **position, size, orientation, or shape** of an object in a coordinate system.
3. The primary goal is to manipulate graphical objects to simulate **motion (Animation)**, change the **Viewing**, or create/modify objects for **modeling** within a computer graphics scene.



Types of transformation

- **Geometric Transformations:** Change the position, size, or shape of an object within the scene's coordinate system (World Space). They include **Translation, Rotation, Scaling, Reflection, and Shearing**.
- **Viewing Transformations:** Operations that map the coordinates of objects from the scene space (World Space) to the display space (Screen Space). The **Window-to-Viewport Transformation** is the prime example.

	Geometric	Viewing
Goal	To change the object's properties (position, size, shape, orientation).	To map a part of the scene onto a specified area of the display device.
Action	Transforms the object itself.	Transforms the coordinate system (or the perspective).
Coordinate Space	Occurs within World Space or Modeling Space .	Maps coordinates from World Space to Screen Space (Viewport).
Key Operations	Translation, Rotation, Scaling, Reflection, Shearing.	Window-to-Viewport Mapping (Normalization, scaling, and translation to screen coordinates).
Effect on Scene	Defines the object's location and appearance in the virtual world.	Defines what is seen (the window) and where it is displayed (the viewport).
Analogy	Moving/Reshaping a model on a stage.	Aiming a camera at the stage and choosing the frame size.

Basic of 2D Transformations

- **Translation** – moves the object.
- **Scaling** – changes the size.
- **Rotation** – changes the orientation.
- **Reflection** – flips the object.
- **Shearing** – distorts the shape
- **Translation**
 - Definition: The process of moving an object from one location to another without changing its shape or orientation. It is defined by a translation vector (T_x, T_y) .
 - Formula
 - $X' = X + T_x$
 - $Y' = Y + T_y$

Scaling

- Definition: The process of changing the size of an object (enlarging or shrinking) relative to a reference point (usually the origin). It is defined by the scaling factors (S_X, S_Y).
- Formula
 - $X' = X \times S_X$
 - $Y' = Y \times S_Y$
- Types of Scaling:
 - Uniform Scaling: Occurs when $S_X = S_Y$ The object's size changes while its aspect ratio is preserved, meaning its shape is not distorted.
 - Non-uniform Scaling: Occurs when $S_X \neq S_Y$.This stretches or compresses the object differently along the x and y axes, resulting in shape distortion.

Rotation

- Definition: The process of changing the orientation of an object about a fixed reference point. The origin (0,0) is often used as the pivot point, and the rotation is performed by an angle θ . (A positive angle usually denotes counter-clockwise rotation).
- Formula (Rotation about the Origin):
 - $X' = X \cos \theta - Y \sin \theta$
 - $Y' = X \sin \theta + Y \cos \theta$

Example:-

A triangle has vertices at: $A(1,1), B(3,1), C(2,4)$

Apply the following sequence of transformations:

Scale the triangle by $S_x=2, S_y=0.5$ about the origin.

Rotate the resulting triangle by 90° counterclockwise about the origin.

Translate it by $T_x=3, T_y=2$.

Find the final coordinates of the transformed triangle.

Matrix Representation

- All 2D transformations can be represented as **3×3 matrices**, which makes it easy to combine multiple transformations using matrix multiplication.

- **Translation:**

- $$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

- **Scaling:**

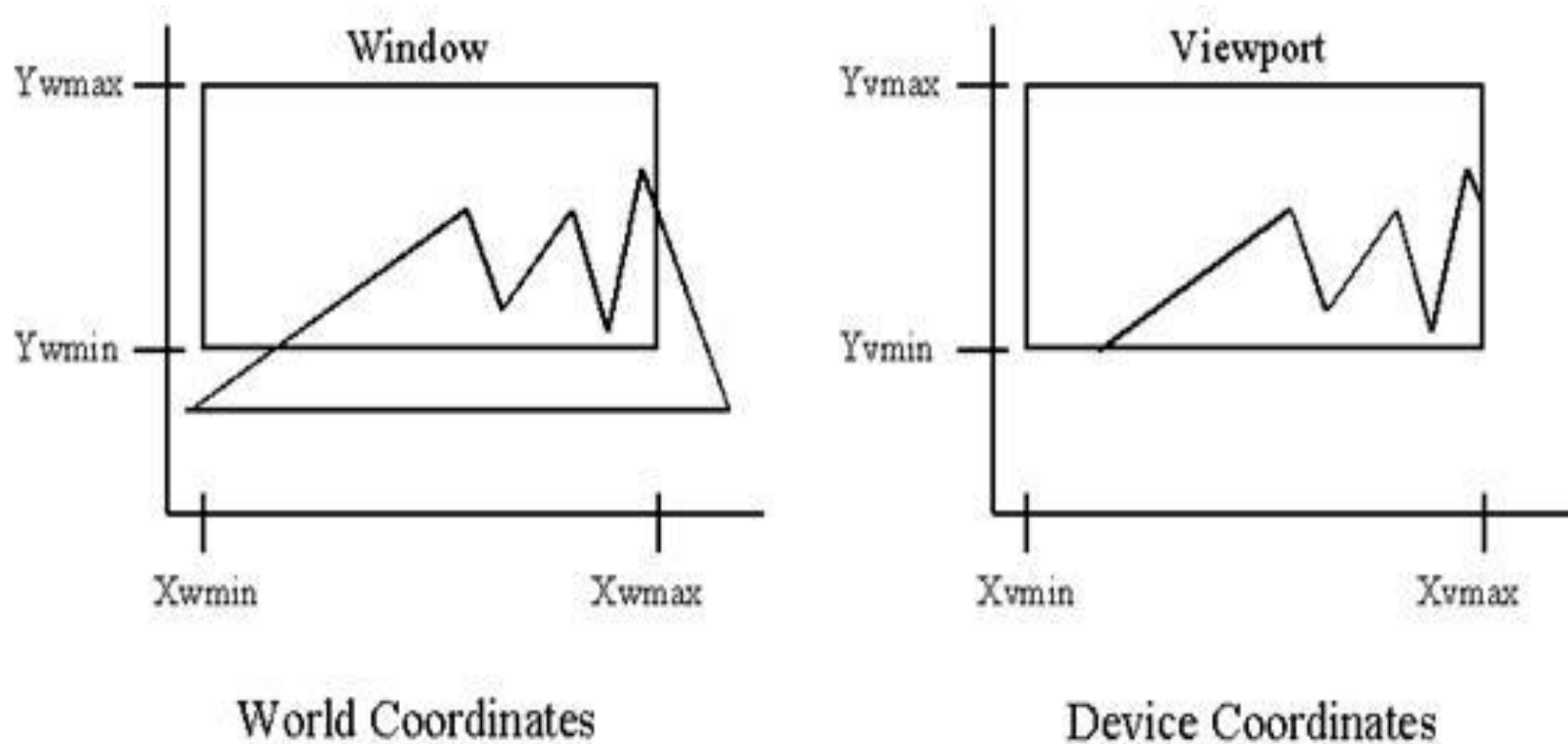
- $$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

- **Rotation (around origin):**

- $$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Windows to Viewport Transformation

The Window-to-Viewport Transformation is a crucial step in computer graphics viewing pipelines. It's the process of **mapping** a designated rectangular region of the virtual scene (**the Window**) onto a specified rectangular area of the physical display screen (**the Viewport**).



Key points

Term	Meaning
Window	The rectangular region in the world coordinate system that you want to display.
Viewport	The rectangular area on the output device (screen) where the window will appear.
Mapping	The process of converting coordinates from the window to the viewport.

Scaling Factors

To map the window area into the viewport, we need to compute **scaling factors** for both x and y directions:

$$S_x = \frac{x_{vmax} - x_{vmin}}{x_{wmax} - x_{wmin}}$$
$$S_y = \frac{y_{vmax} - y_{vmin}}{y_{wmax} - y_{wmin}}$$

These determine how much to stretch or compress the window content to fit inside the viewport.

Transformation Equations

- Each point (x_w, y_w) in the **window** is mapped to a point (x_v, y_v) in the **viewport** using:
 - $x_v = x_{vmin} + (x_w - x_{wmin}) \cdot S_x$
 - $y_v = y_{vmin} + (y_w - y_{wmin}) \cdot S_y$
- These equations translate and scale the coordinates so that the window content fits exactly into the viewport area.

Given the following data:

- Window coordinates:

$$X_{wmin} = 0, Y_{wmin} = 0$$
$$X_{wmax} = 10, Y_{wmax} = 10$$

- Viewport coordinates:

$$X_{vmin} = 100, Y_{vmin} = 100$$
$$X_{vmax} = 200, Y_{vmax} = 200$$

Transform the following window points to their corresponding viewport coordinates using the window-to-viewport transformation equations:

- 1.(0, 0)
- 2.(5, 5)
- 3.(10, 10)

Answer

- **Window coordinates:**
- $Xw_{min} = 0, Yw_{min} = 0, Xw_{max} = 10, Yw_{max} = 10$
- **Viewport coordinates:**
- $Xv_{min} = 100, Yv_{min} = 100, Xv_{max} = 200, Yv_{max} = 200$

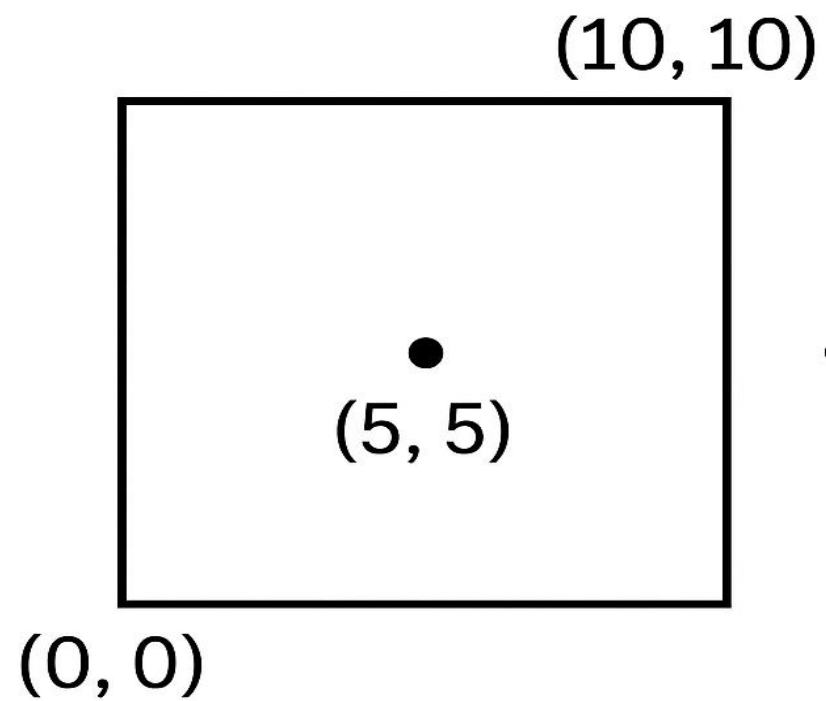
1-Find the scaling factors

- $Sx = \frac{Xv_{max} - Xv_{min}}{Xw_{max} - Xw_{min}} = \frac{200 - 100}{10 - 0} = \frac{100}{10} = 10$
- $Sy = \frac{Yv_{max} - Yv_{min}}{Yw_{max} - Yw_{min}} = \frac{200 - 100}{10 - 0} = \frac{100}{10} = 10$

2- Apply the transformation equations

- $Xv = Xv_{min} + (Xw - Xw_{min}) \times Sx$
- $Yv = Yv_{min} + (Yw - Yw_{min}) \times Sy$
- **For point (0, 0):**
- $Xv = 100 + (0 - 0) \times 10 = 100$
- $Yv = 100 + (0 - 0) \times 10 = 100$
- **Transformed point = (100, 100)**

Window



Viewport

